

Proportional DC to HV DC Converters

0 to + or - 100 through 0 to + or - 12,000 VDC up to 3 Watts
E Series



FEATURES

- Low Ripple
- Low EMI/RFI
- Proportional Input/Output
- Exceptional Reliability
- Compact, PCB Mount Package

OPTIONS

RoHS(-'R' suffix denotes the product is designed to meet RoHS requirements i.e E01R)
External Mounting Box, See AB Series
Output Center Tap, See CT Series
Mounting Holes
Alternate Pin Patterns Available
Epoxy: **A.** Low Outgassing (NASA approved per ASTM E-595-93)
B. UL 94 V0 flammability rating

APPLICATIONS

- Electro-Static Field Generation
- HV Op Amps
- Grid Bias
- Spectrometry
- Piezo Devices
- Lamp Ignition
- Q Switches
- Ion Pumps
- Capacitor Charging
- Electrophoresis
- Image Intensifiers
- Printers

PHYSICAL CHARACTERISTICS

SIZE: 1.5 x 2.5 x 0.85 (38 x 64 x 22)
WEIGHT: 3 Ounces (85 grams) Approx.
PACKAGING: Epoxy Encapsulated
CASE MATERIAL: Glass-filled Epoxy
PINS: .031 (.79) Dia., .20 (5.1) Min. Length

ELECTRICAL SPECIFICATIONS

INPUT VOLTAGE: See Table
TYPICAL TURN-ON VOLTAGE: 0.7 Volts
OUTPUT VOLTAGE: See Table
OUTPUT CURRENT: See Table
RIPPLE: See Table
ISOLATION: 3,500 Volts +Vout
E70 – E121: 500V + Vout
EFFICIENCY: >60% Typical
OPERATING TEMP: -10° to +60° C
E70 – E121: -10° to +50° C

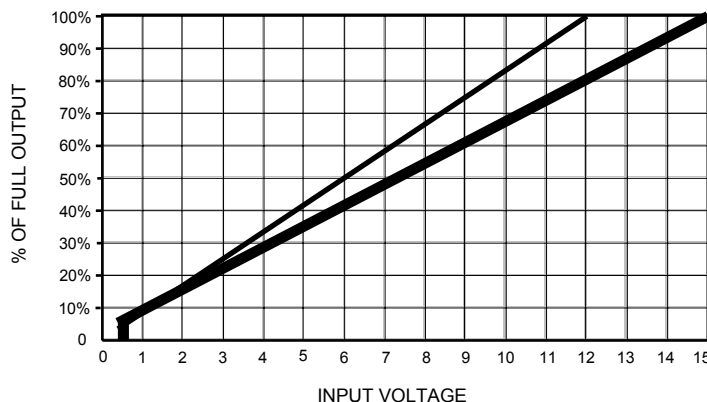
The E Series is a broad line of small, versatile, component level building blocks that provide up to 12,000 VDC at up to 3 watts continuous output power*² in a PCB mount package. This series features low ripple, noise, and EMI/RFI by utilizing a quasi-sinewave oscillator, excellent filtering techniques and a fully enclosed pot core transformer. The output voltage is

directly proportional to the input voltage, and is linear from approximately 0.7 volts to maximum input. The isolated output allows for user selectable output polarity. Options include mounting holes, external aluminum box, and an output center tap which, when grounded, provides both positive and negative outputs from one low cost module.

MODEL	INPUT CURRENT (NO LOAD)	INPUT CURRENT (FULL LOAD)	INPUT VOLTAGE	OUTPUT* ³ VOLTAGE	OUTPUT* ² CURRENT	RIPPLE P-P
E01	<175 mA	<425 mA	0 to 12	0 to +/-100	30 mA	0.1%
E02	<125 mA	<400 mA	0 to 12	0 to +/-200	15 mA	0.25%
E02-5	<125 mA	<400 mA	0 to 12	0 to +/-250	12 mA	0.75%
E03	<125 mA	<400 mA	0 to 12	0 to +/-300	10 mA	1.0%
E05	<150 mA	<400 mA	0 to 12	0 to +/-500	6 mA	0.02%
E06	<175 mA	<400 mA	0 to 12	0 to +/-600	5 mA	0.1%
E08	<125 mA	<400 mA	0 to 12	0 to +/-800	3.75 mA	0.03%
E10	<125 mA	<400 mA	0 to 12	0 to +/-1,000	3 mA	0.03%
E15	<150 mA	<400 mA	0 to 12	0 to +/-1,500	2 mA	0.05%
E20	<175 mA	<425 mA	0 to 12	0 to +/-2,000	1.5 mA	0.25%
E30	<175 mA	<400 mA	0 to 15	0 to +/-3,000	1 mA	0.25%
E40	<175 mA	<400 mA	0 to 15	0 to +/-4,000	0.75 mA	0.5%
E50	<175 mA	<400 mA	0 to 15	0 to +/-5,000	0.6 mA	0.5%
E60	<175 mA	<400 mA	0 to 15	0 to +/-6,000	0.5 mA	0.5%
E70* ¹	<175 mA	<350 mA	0 to 15	0 to +/-7,000	0.43 mA	1%
E80* ¹	<175 mA	<300 mA	0 to 15	0 to +/-8,000	0.25 mA	1.25%
E101* ¹	<175 mA	<300 mA	0 to 15	0 to +/-10,000	0.2 mA	1.5%
E121* ¹	<175 mA	<300 mA	0 to 15	0 to +/-12,000	0.16 mA	1.5%

* Note 1. This unit has flying leads on the output and clearance holes for mounting
2. At Maximum Rated Output Voltage.
3. Output Voltage is load dependent. Under light or no load conditions, reduce input voltage so maximum rated output voltage is not exceeded.

TYPICAL INPUT vs. OUTPUT VOLTAGES



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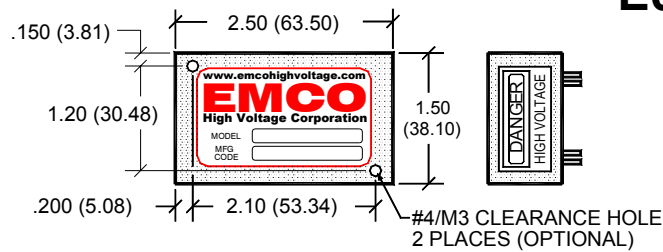
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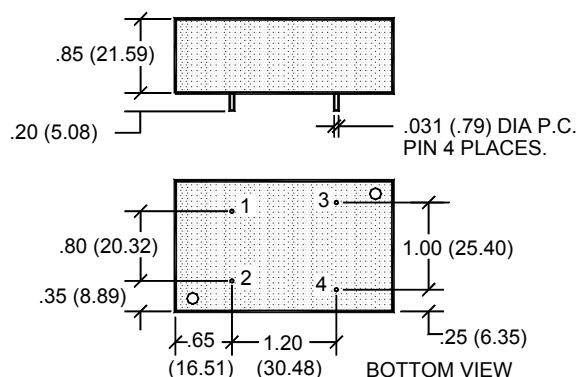
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E Series

E01-E60

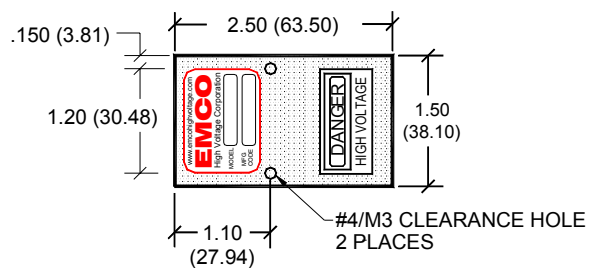


Pin #	Function
1	(+) Input
2	(-) Input
3	(+) Output
4	(-) Output

Dimensions are in Inches
Dimensional Tolerances: $\pm .03$ ($\pm .76$)
(Metric Equivalents in Parentheses)



E70-E121



Pin #	Function
1	(+) Input
2	(-) Input
3*	(+) Output
4*	(-) Output

*Output Leads

Dimensions are in Inches
Dimensional Tolerances: $\pm .03$ ($\pm .76$)
(Metric Equivalents in Parentheses)

